5.3 European Union

Key highlights

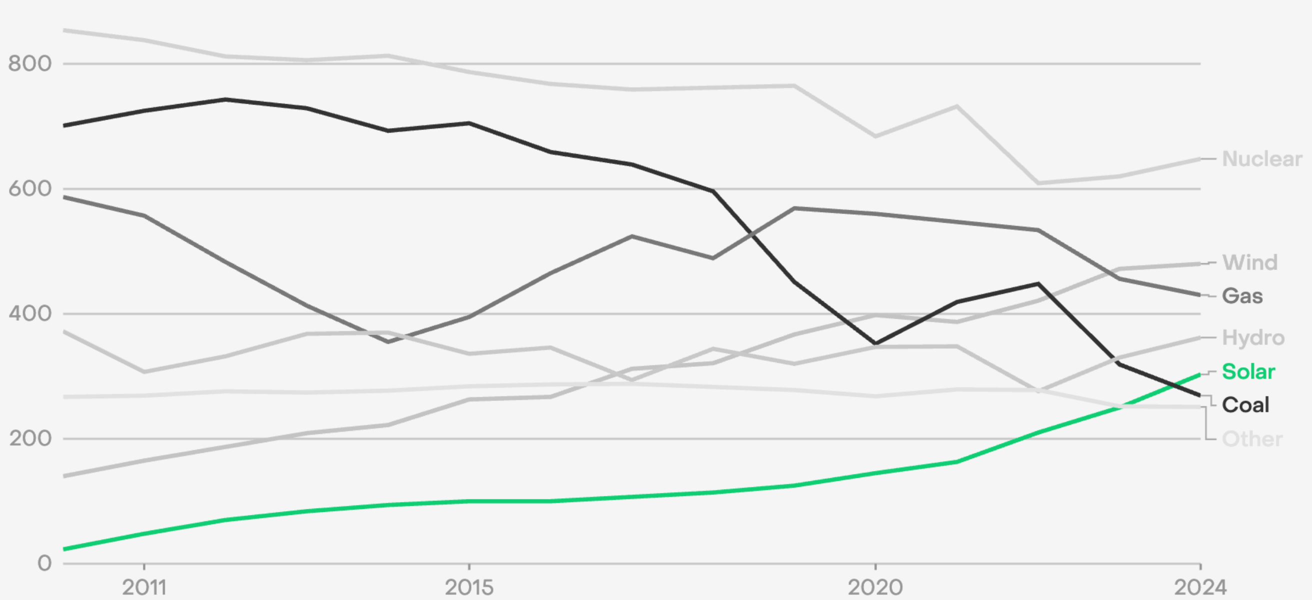
01 **Solar power overtook coal generation in the EU for the first time in 2024**

02 **In 2024, the EU saw the largest fall in coal generation globally**

03 **The EU's biggest two generation sources – nuclear and wind – are both low-carbon, with gas and coal in third and sixth place respectively**

Solar overtook coal generation in the EU for the first time in 2024

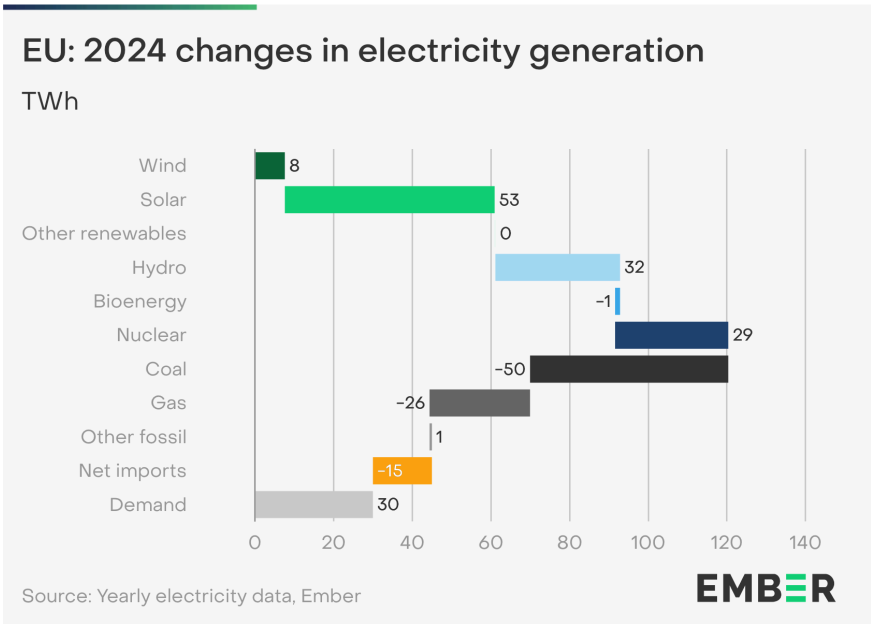
Electricity generation (TWh)



Source: Yearly electricity data, Ember
'Other' includes bioenergy, other fossil and other renewables

EMBER

In 2024, solar was the European Union's fastest-growing power source, with a 21% increase (+53 TWh) compared to 2023. This rise accounted for 11% of the global increase in solar generation. The EU saw a record amount of new capacity additions in 2024, driving this solar increase despite lower solar radiation in 2024 than in 2023.



Wind generation grew by 8 TWh year-on-year in 2024. This growth is lower than the average 30 TWh year-on-year increase seen between 2019 and 2023. While capacity additions continued in 2024, wind conditions were less favourable than in 2023, leading to lower than expected generation.

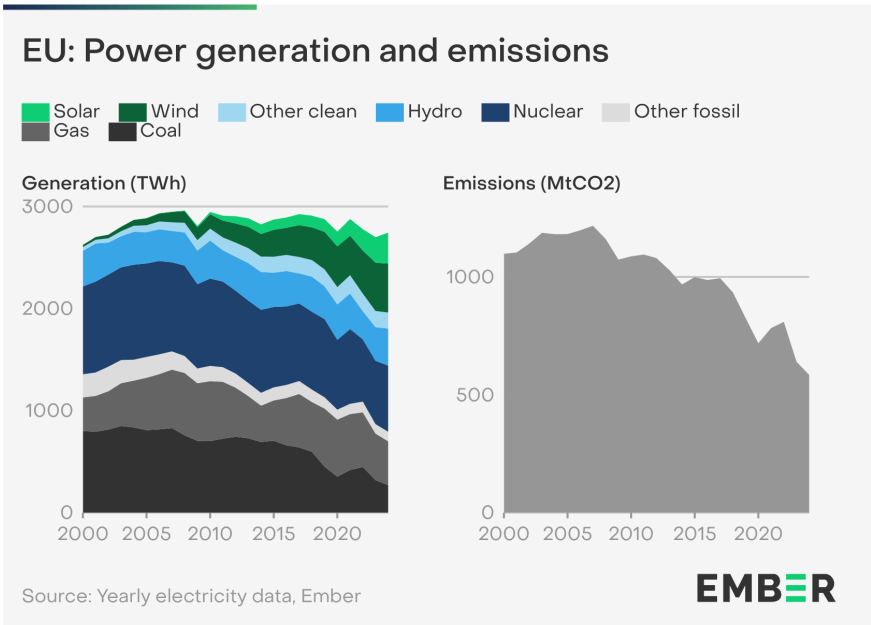
Hydro and nuclear generation increased by 32 TWh (+9.6%) and 29 TWh (+4.6%) respectively. Hydro increased due to favorable rainfall across most of Europe, while nuclear power was boosted because of reduced downtime in France.

Fossil gas generation fell for the fifth year in a row (-26 TWh, -5.6%). Coal generation in the EU fell by 50 TWh (-16%), continuing a second consecutive year of decline. This was the largest decline in any power sector globally.

EU demand rose by 30 TWh (+1.1%), steadying after large falls in 2022 and 2023.

The EU has proven that a deep transformation of the power sector is achievable and beneficial. In 2024, EU power sector emissions were down to 585 million tonnes of CO_2 ($MtCO_2$), below half of their 2007 peak.

Over the last five years, coal generation has fallen by 182 TWh (-40%), with Austria, Sweden and Portugal phasing it out completely, while countries with large coal fleets like Germany saw significant plant closures. At the same time, gas generation has decreased in each of the last five years, and was 139 TWh (-24%) lower in 2024 than it was in 2019. The reduction in gas generation has enhanced the bloc's energy security amidst Russia's invasion of Ukraine and the gas price volatility seen over the last three years. EU fossil generation is now at its lowest level for more than forty years (793 TWh).



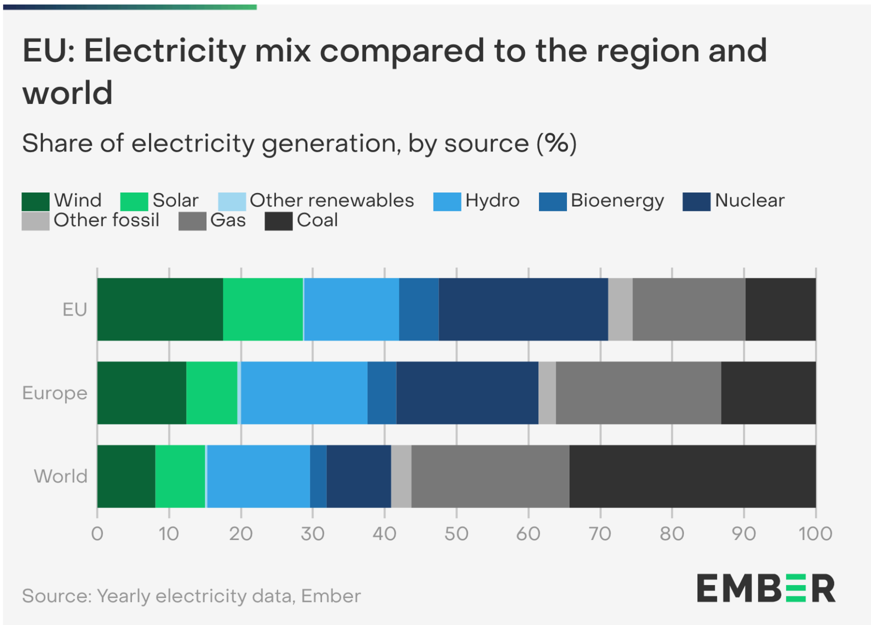
The key driver of this has been a significant rise in wind and solar generation. Wind and solar's share in the EU power mix has increased from 17% in 2019 to 29% in 2024, with wind generation increasing by 113 TWh (+31%) and solar generation by 179 TWh (+143%).

Hydropower capacity has remained unchanged over the last five years, with changes in generation dominated by weather conditions. Meanwhile, nuclear capacity has decreased from 110 GW to 96 GW, although the largest changes in generation have come from outages and maintenance.

As a result of its power sector transformation, the EU has cemented its position as a global leader in clean power, showing it is possible to integrate high shares of variable renewables. 71% of the region's electricity came from clean sources in 2024, far above the global average of 41%. The EU's biggest two generation sources – nuclear and wind – are both low-carbon, with gas and coal in third and sixth place respectively.

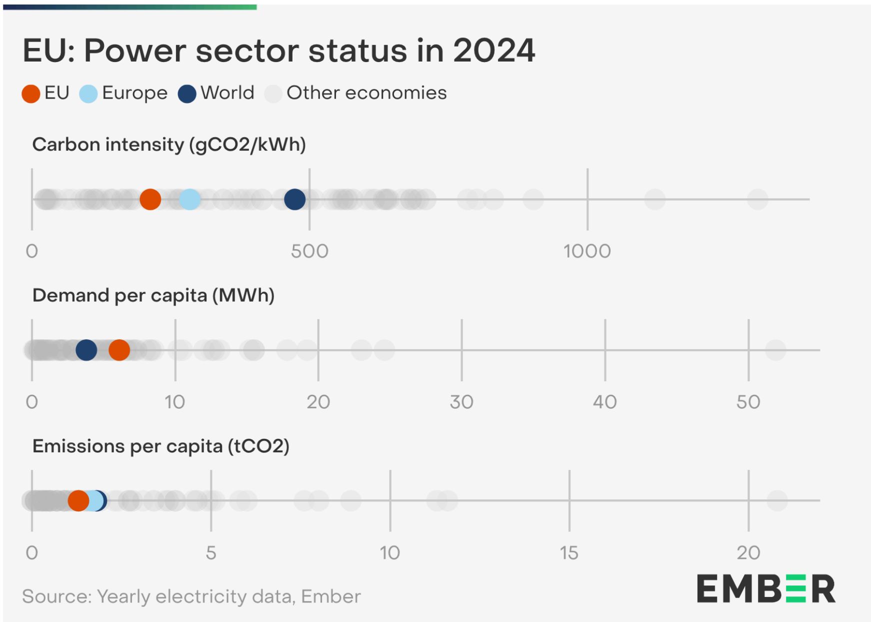
The EU obtained 18% of its electricity from wind power, more than double the global average. Solar provided 11% of the EU's electricity generation in 2024, surpassing coal for the first time ever. Out of the 15 countries with the highest solar shares in 2024, seven were EU Member States. France's sizeable nuclear fleet meant that the bloc generated 24% of its electricity from nuclear power, above the global average.

Fossil generation provided 29% of the EU's electricity, half of the global average of 59%. The main difference was a much lower share of coal generation. Coal generation fell to below 10% (9.8%) of total EU electricity generation, compared to the global average of 34%.



The carbon intensity of electricity generation in the EU was 213 gCO2 per kWh in 2024, less than half the global average (473 gCO2/kWh).

The EU's per capita power demand was 6.1 MWh, 60% higher than the world average and half the demand per capita of the United States.



The low CO2 intensity and demand per capita meant that the EU's per capita power sector emissions were 1.3 tonnes of CO2 (tCO2), 28% less than the world average of 1.8 tCO2 and about one third of those in China and the United States.

5.4 India

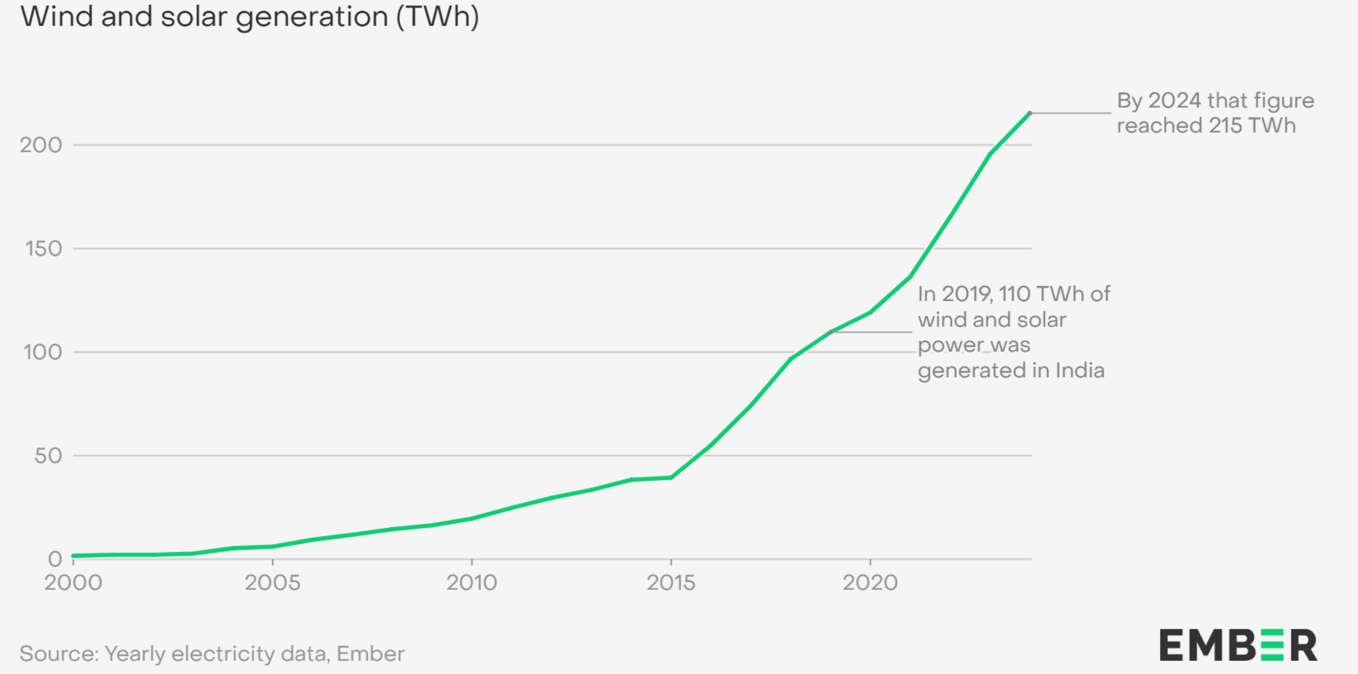
Key highlights

01 **India's wind and solar generation doubled in the five years leading to 2024**

02 **India overtook Germany to become the world's third largest generator of electricity from wind and solar in 2024**

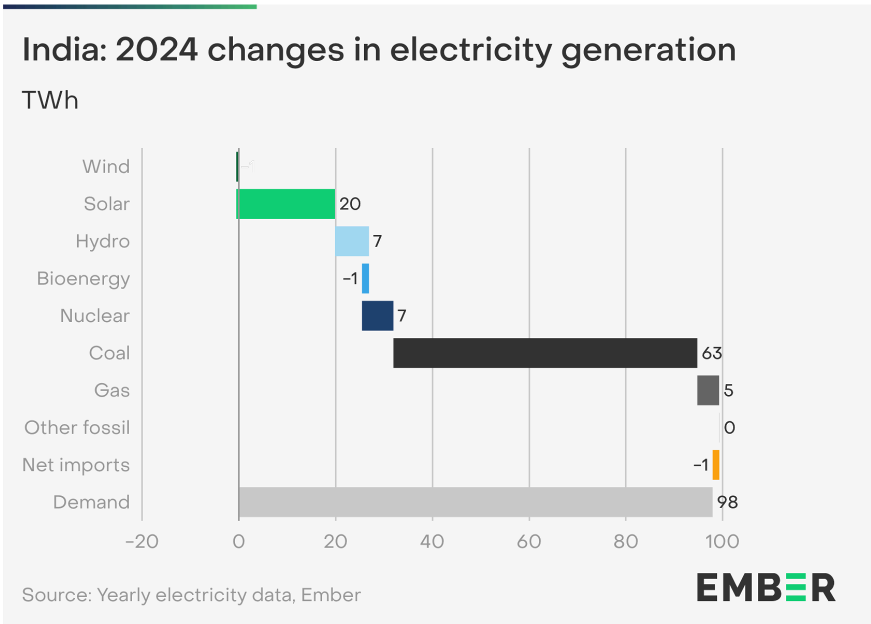
03 **Coal generation growth met 64% of India's electricity demand growth in 2024, a substantial decline from the 91% in 2023**

India's wind and solar generation shows no signs of slowing down - nearly doubling in the last five years



Source: Yearly electricity data, Ember

India's power demand increased by 5% (98 TWh) in 2024, slightly less than the 7% growth in 2023, and in line with the country's average annual demand growth rate for the last decade (+5.5%). India had the third-largest demand increase in the world.



Clean generation increased by 32 TWh (+7.4%) in 2024, meeting 33% of India's demand increase. The clean generation increase was in part due to a rebound of hydro (+7 TWh, +4.7%) after a decline of 26 TWh in 2023.

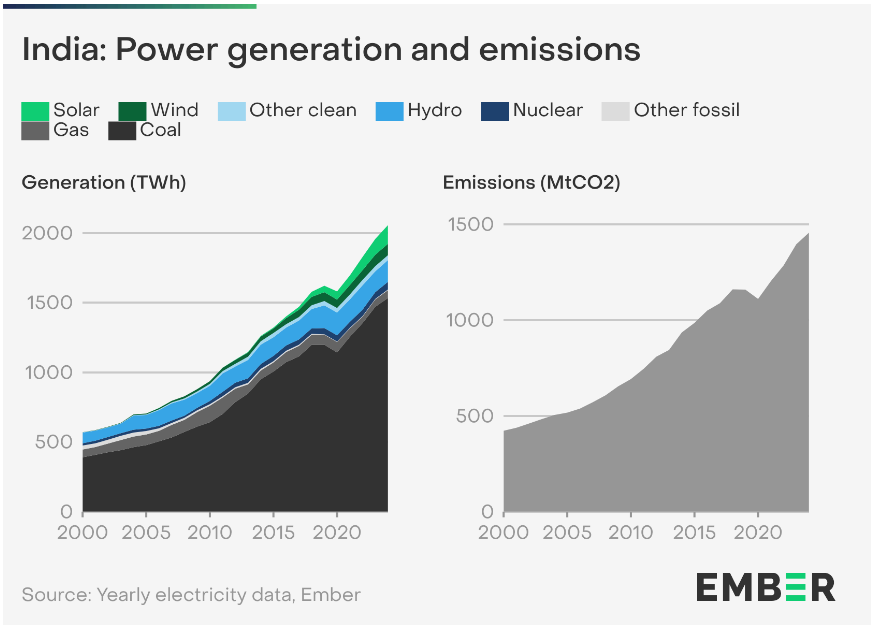
India recorded the fourth-largest increase in solar generation of any country in 2024 at 20 TWh (+18%), more than the total solar generation of the United Kingdom. This was driven by record capacity additions in 2024, more than double the additions in 2023. However, lower solar radiation meant that the generation increase seen in 2024 did not fully reflect the impact of this capacity boom.

Fossil generation increased by 67 TWh (+4.4%) in 2024, significantly lower than 2023's figure of 124 TWh (+8.8%). Coal generation grew by 4.3% in 2024, almost half the average of the prior three years (8.8%). Coal generation growth met 64% of India's electricity demand growth in 2024, a substantial decline from the 91% in 2023.

India's demand had been rising by more than 7% every year since the Covid-19 decline of 2020, but in 2024, demand growth fell back to the historic ten-year trend of around 5%.

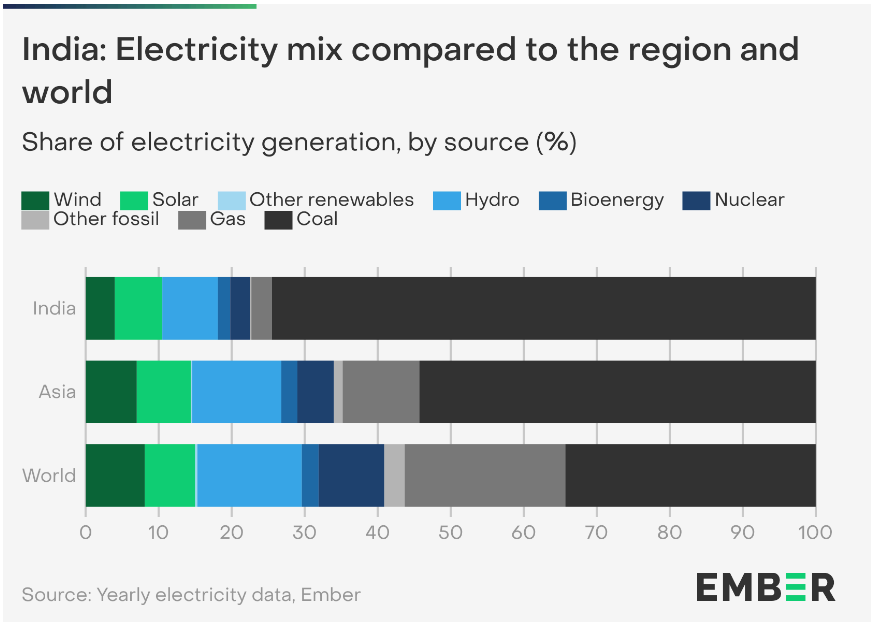
India's wind and solar generation almost doubled over the five years leading to 2024, from 110 TWh to 215 TWh. As a result, India overtook Germany in 2024 to become the world's third largest generator of electricity from wind and solar. Hydro generation in 2024 was roughly the same as five years ago, and nuclear power increased by 10 TWh (+21%) over the same period.

India's coal-fired power generation also continued to rise, almost doubling from 2012 (787 TWh) to 2024 (1,534 TWh). In 2018, India overtook the US to become the second-largest coal generator, and now has more than twice the coal generation of the US.



As a result, power sector emissions continue to rise, reaching 1,457 MtCO2 in 2024. This makes India the world's third-largest power sector emitter, although emissions per capita remain well below the global average.

India is one of only ten countries that is currently planning for a tripling of renewables capacity by 2030 (from 2022), with the goal of reaching 500 GW of clean power capacity. By October 2024, the country had achieved [200 GW of RE capacity.] India is focused on domestic policies that support solar modules, batteries, and electrolysers to support the energy transition, while meeting development milestones such as the recent delivery of 100% electricity access.

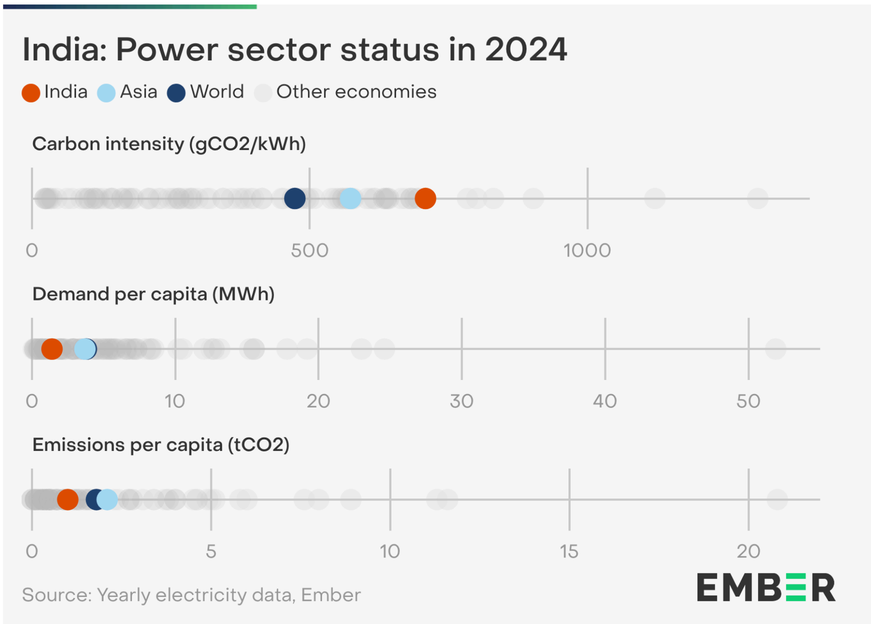


India generated 78% of its electricity from fossil sources in 2024, primarily driven by coal at 75%. This is above the 66% regional average for Asia and well above the 59% global average.

Clean generation made up 22% of India's electricity mix in 2024. Hydro was India's largest clean electricity source at 8%. Solar and wind combined reached 10% in 2024. However, this is still below the global average (15%) and China (18%).

The carbon intensity of India's power sector was 708 gCO2/kWh in 2024. This is higher than both the average in Asia of 573 gCO2/kWh and the global average of 473 gCO2/kWh.

Despite having the third-highest demand of any country, the electricity demand per capita was 1.4 MWh, less than half of the Asian regional average of 3.7 MWh and the global average of 3.8 MWh.



India's electricity is heavily coal-reliant. However, the low per capita electricity use meant that India's per capita emissions were 1 tCO2, much lower than the global average (1.8 tCO2).

5.5 Russia

Key highlights

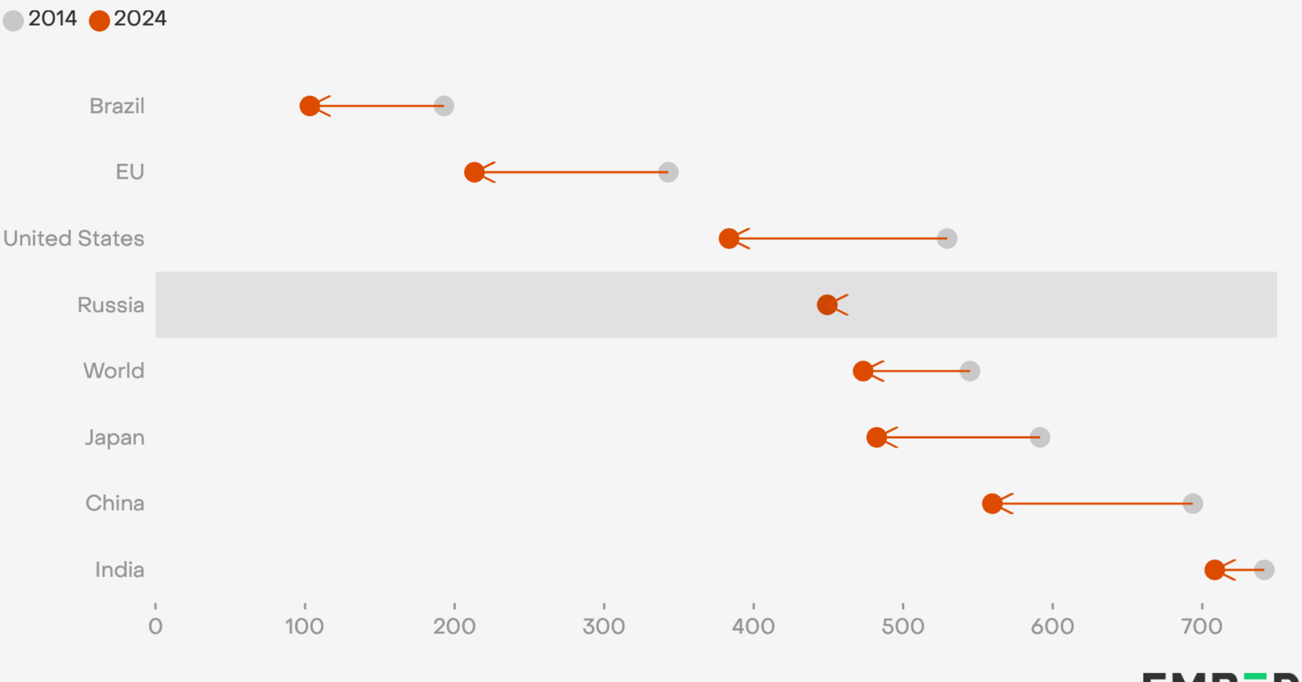
01 **Russia set a new record high for both coal and gas generation in 2024**

02 **Wind and solar accounted for less than 1% of Russia's electricity mix, the second-lowest share in the G20**

03 **The carbon intensity of Russia's electricity generation in 2024 remained unchanged from a decade ago**

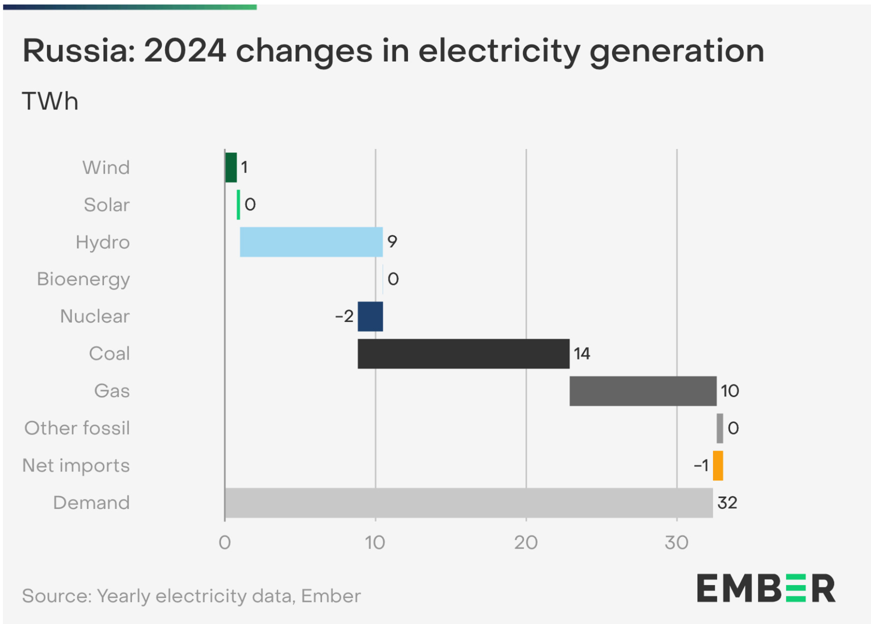
Russia's carbon intensity of electricity fails to improve, while the rest of the world makes strides

Carbon intensity of electricity generation (gCO2/kWh)



Source: Yearly electricity data, Ember

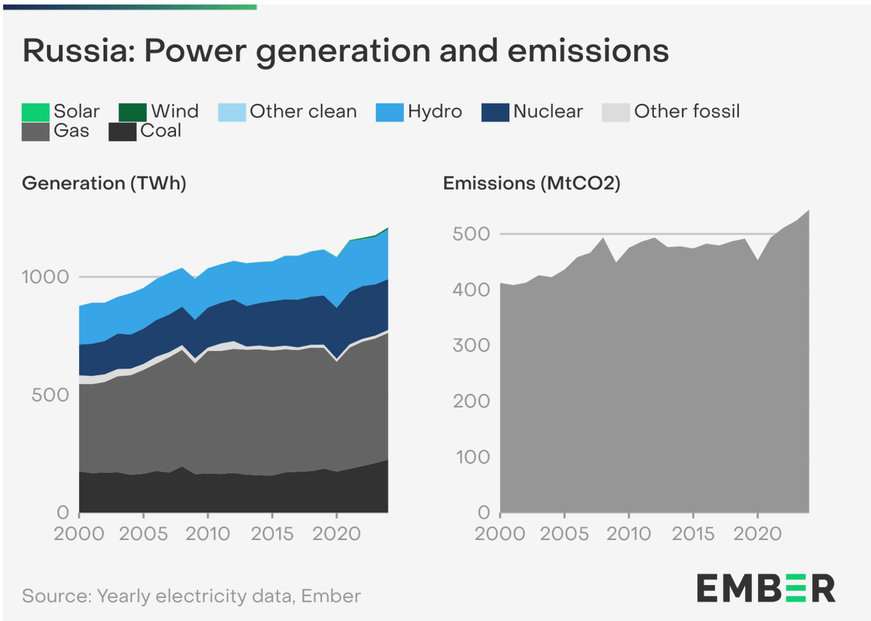
Demand in Russia grew by 32 TWh (+2.8%) in 2024, lower than the global demand increase of 4%. Excluding the 5.7% rise in 2021, following the Covid-19 pandemic, this was Russia's highest demand increase since 2010. Fossil generation met most of this increase, with a 14 TWh (+6.7%) coal rise and a gas increase of 9.8 TWh (+1.9%).



Hydro increased by 9.5 TWh (+4.7%) in 2024, following a 3.2 TWh (+1.6%) increase in 2023. Nuclear saw a small decline of 1.7 TWh (-0.8%) in 2024.

Wind and solar only play a small role in Russian generation. Wind saw a 0.8 TWh increase and solar generation grew by 0.2 TWh.

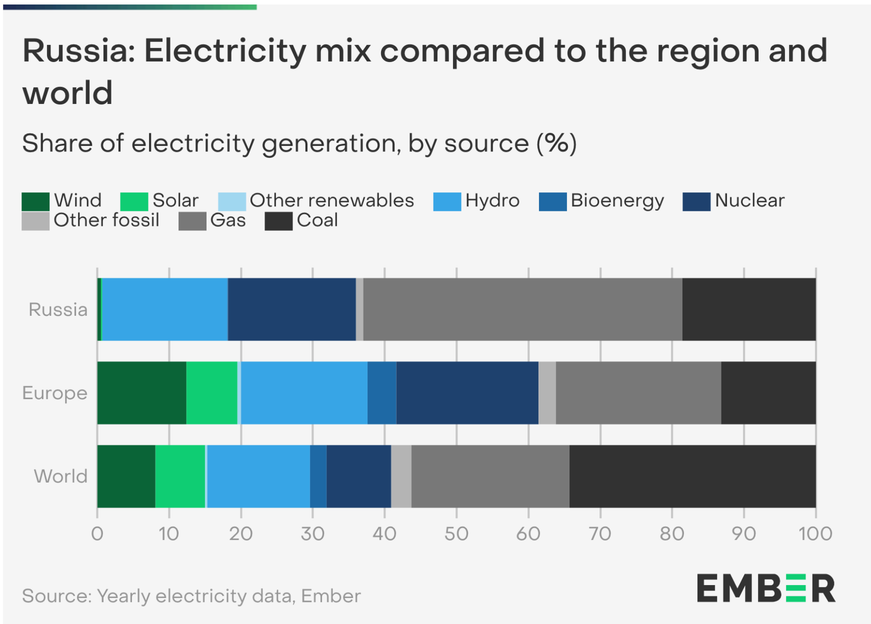
Russia set a new record for both coal and gas generation in 2024, with fossil generation continuing to meet growing electricity demand. As a result, Russia was the world's fourth-largest power sector emitter with a total of 544 million tonnes of CO2 (MtCO2) in 2024.



Power sector emissions in Russia had been relatively flat from 2013 until they dropped due to Covid-19 restrictions in 2020, but have been on a steady increase since then. Emissions were up by 20 MtCO2 in 2024, an increase of 3.8% over 2023, more than double the global average.

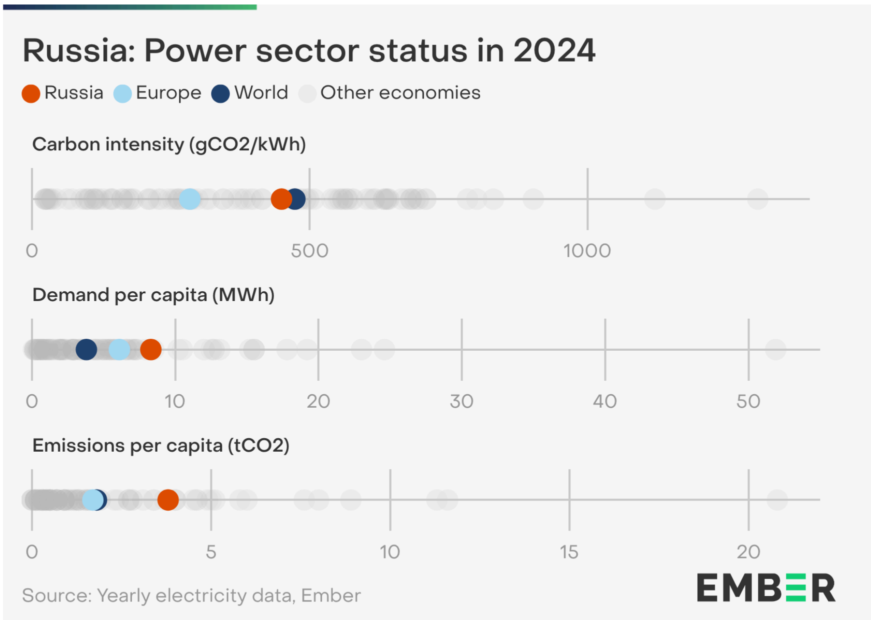
Nuclear and hydro generation have been rising in Russia. In 2024, nuclear generation was 10% higher than in 2015, though falling slightly in market share. Partly due to higher rainfall, hydro generation was 25% higher in 2024 compared to 2015. Solar and wind remained a tiny part of the electricity mix.

Russia has not started its clean energy transition. Its share of wind and solar generation combined was less than 1% of its total power mix, the second-lowest in the G20. Russia lags well behind the European average (20%), China (18%), and the global average of 15%.



Fossil fuels made up 64% of Russia's electricity mix in 2024. Gas made up the largest share at 44%, followed by coal which accounted for 19% of the country's electricity generation. Clean power made up 36% of the electricity mix with nearly all of it coming from hydro and nuclear, contributing 17% and 18% respectively.

Russia's carbon intensity of electricity was 449 gCO2/kWh in 2024, just below the global average of 473 gCO2/kWh.



Russia's demand per capita was 8.3 MWh, more than twice the global average of 3.8 MWh.

As a result of the relatively high demand, Russia's power sector emissions per capita in 2024 were 3.8 tCO2, more than double the global average of 1.8 tCO2.

5.6 Japan

Key highlights

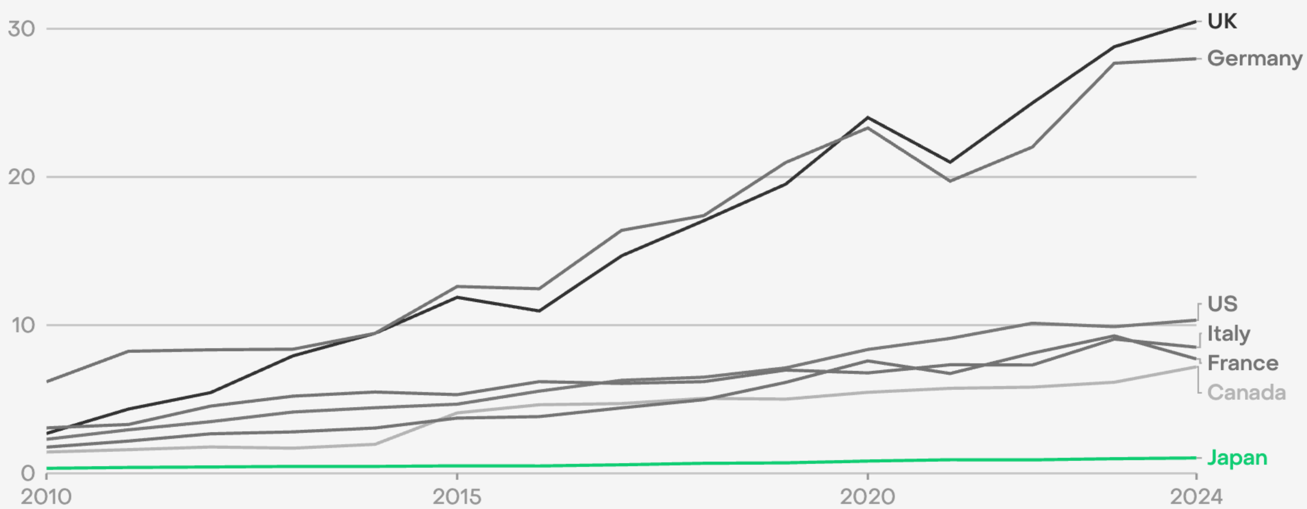
01 **Wind power has grown significantly in all G7 countries except Japan**

02 **Japan remained the fourth-largest solar generating country globally**

03 **Japan's power sector emissions fell by 1.4% in 2024**

The G7 has seen strong wind growth since 2010, but Japan is falling behind

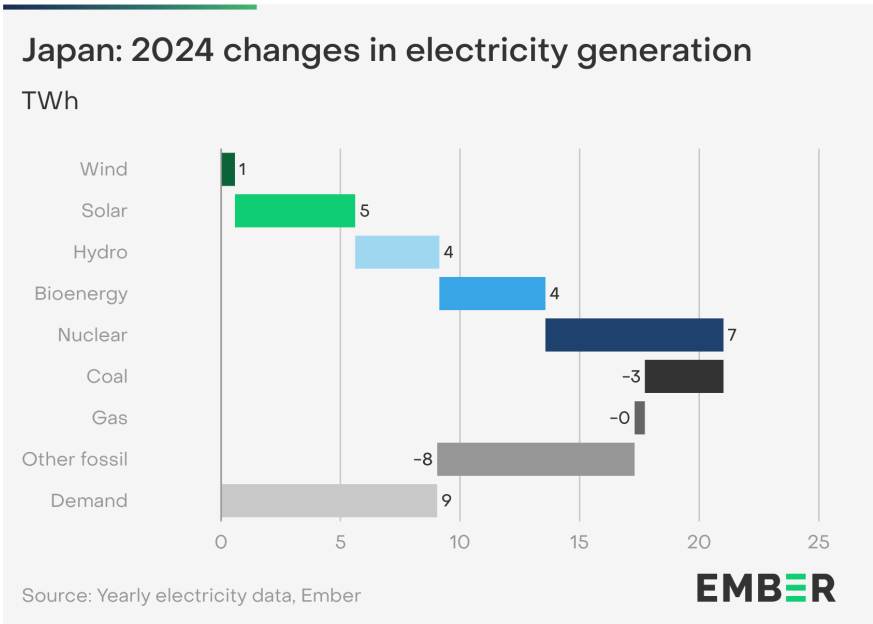
Share of wind in electricity generation (%)



Source: Yearly electricity data, Ember

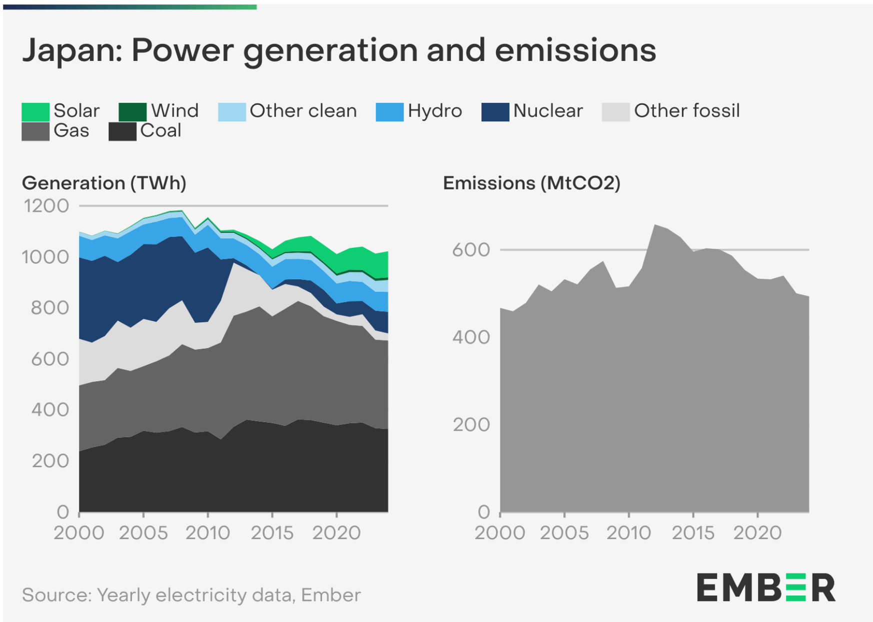
EMBER

In 2024, Japan's power demand increased by 9 TWh (+0.9%), in part as a result of hotter temperatures in summer months. Fossil generation fell by 12 TWh (-1.7%). This was much lower than the fall in fossil generation in 2023 (-63 TWh, -8%).



Japan's clean generation growth was led by an increase in nuclear generation (+7.5 TWh, +9.6%) and solar (+5 TWh, +5.2%). Overall clean generation was up by 21 TWh (+7%), just over half of the increase in 2023 (+36 TWh; +13.6%).

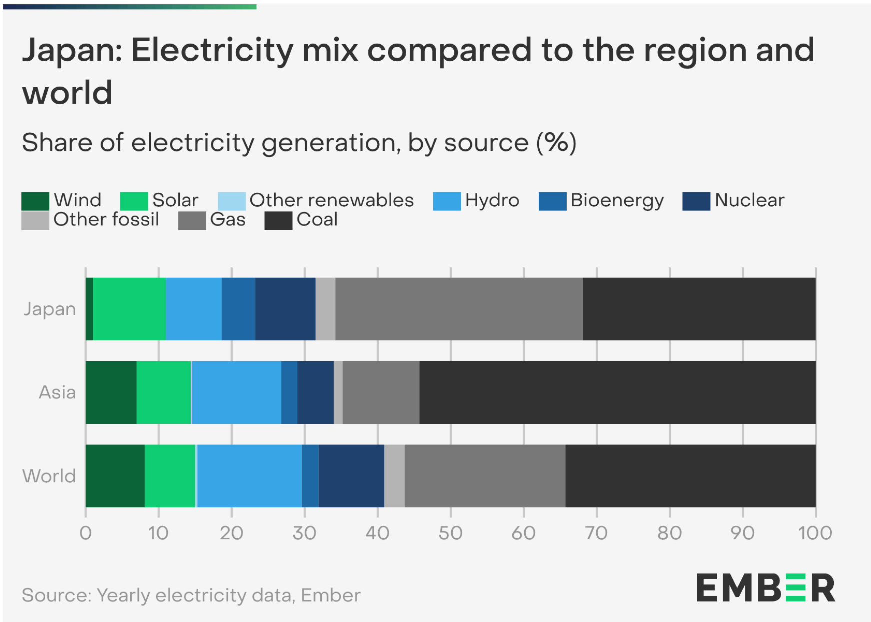
Japan's power sector emissions peaked in 2012 and have declined 25% since then, reaching 493 million tonnes of CO2 (MtCO2) in 2024. This was the lowest value in the past 22 years, but still slightly higher than in 2000. This is because coal and gas still generate around two-thirds of Japan's electricity, though in recent years they have been increasingly replaced by nuclear and solar.



Solar's share in the mix has grown significantly in Japan, rising fivefold from 2% of generation in 2014 to 10% in 2024. On the other hand, wind was just 1% of generation in 2024. This is in stark contrast to the rest of the G7, where wind has risen to an average of 12% of total electricity generation.

Nuclear generation has grown in recent years owing to the reopening of reactors closed following the Fukushima nuclear disaster. Two units reopened in 2024. However, due to the slow recovery of nuclear power and the lack of wind power, Japan produced less clean electricity in 2024 (322 TWh) than it did in 2000 (420 TWh).

Japan's largest source of clean electricity was solar (10%) in 2024. However, despite significant wind potential, its share of wind is only 1%. As a result, the wind and solar share was only 11% in 2024, below the global average (15%), regional average (14%), and its neighbour China (18%).

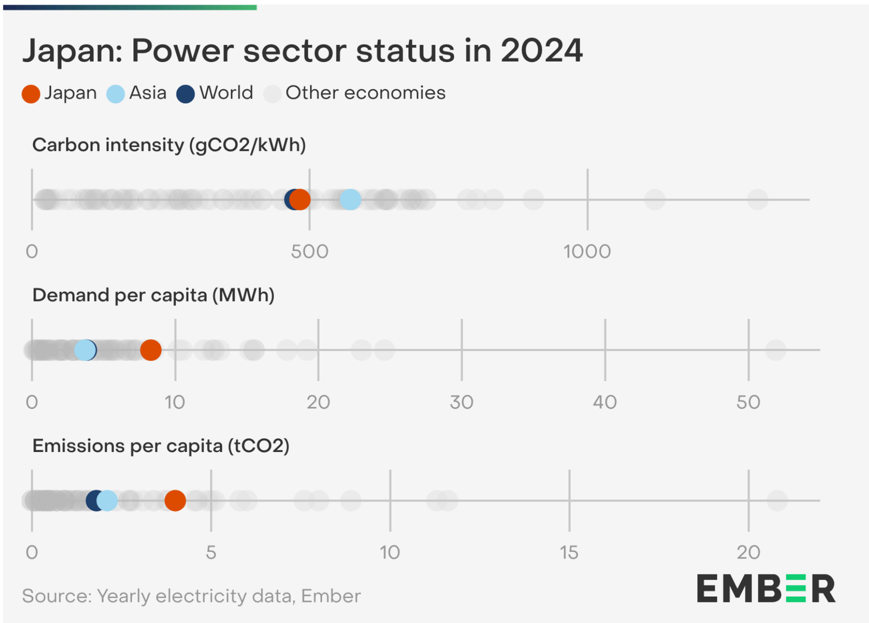


In 2024, Japan relied on fossil fuels for 69% of electricity generation. This is more than the regional average of 66% and global average of 59%.

The share of nuclear in electricity generation was 8.3%, up from 7.6% in 2023 due to the restart of additional nuclear reactors in 2024. There are currently 10 reactors under review for potential restart, and the recent 7th long-term strategic energy plan aims for 20% nuclear share in generation by 2040.

The carbon intensity of Japan's power sector was 482 gCO_2/kWh in 2024, virtually unchanged from 2023, and just above the global average of 473 gCO_2/kWh. Due to the lower contribution of coal in Japan as compared to most of Asia, its carbon intensity was lower than the Asian regional average of 573 gCO_2/kWh.

However, Japan's emissions per capita of 4 tCO_2 were more than twice the global average of 1.8 tCO_2 and almost twice the regional average of 2.1 tCO_2. This is primarily due to Japan's large demand per capita at 8.3 MWh.



5.7 Brazil

Key highlights

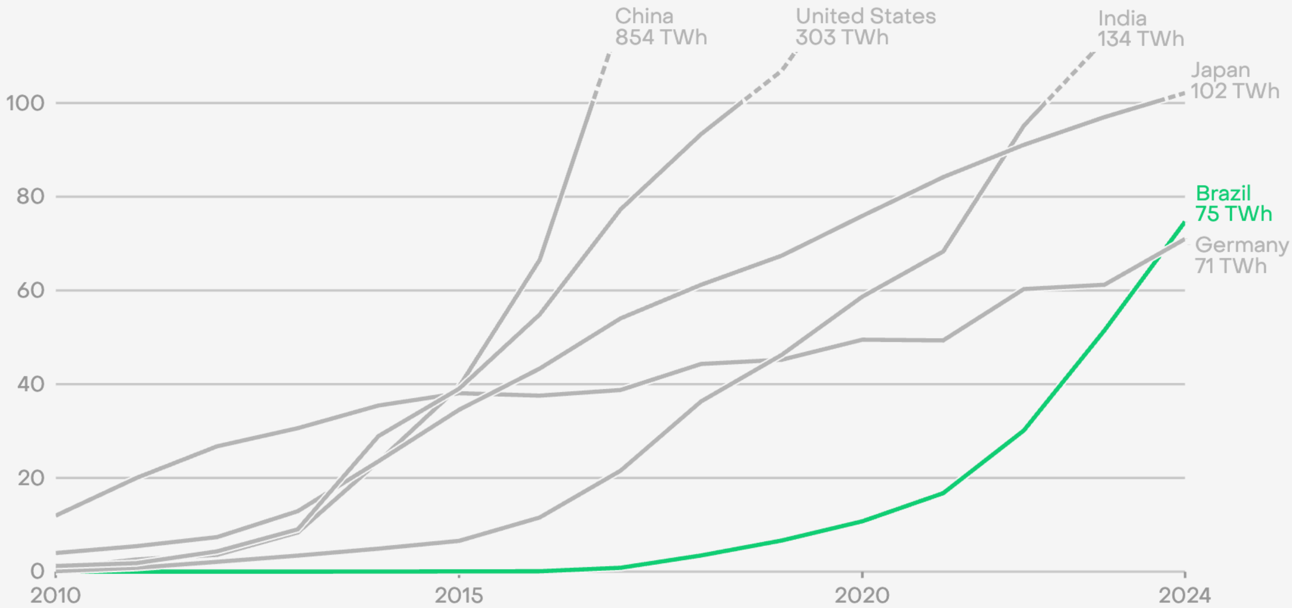
01 **In 2024, Brazil had the world's third-largest increase in both wind and solar generation**

02 **Brazil has overtaken Germany to become the fifth-largest solar generator**

03 **Brazil's power sector emissions peaked a decade ago, and in 2024 were down 32% from 2014**

Brazil overtook Germany to become the fifth largest generator of solar power in 2024

Electricity generation from solar (TWh)

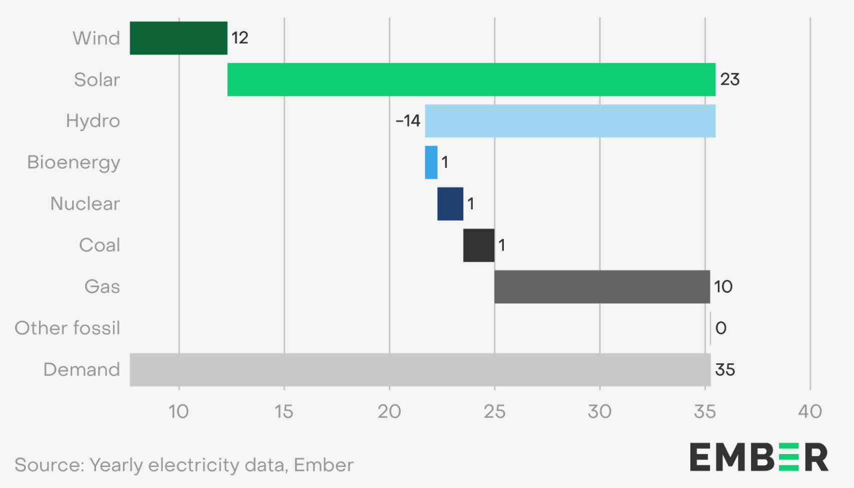


Source: Yearly electricity data, Ember
Electricity generation from solar in 2024 labelled

EMBER

Electricity demand in Brazil was up by 35 TWh (+4.9%) in 2024, similar to the demand increase in 2023 (+35 TWh, +5%). Both years' increases were triple that of 2022 (+11 TWh, +1.6%).

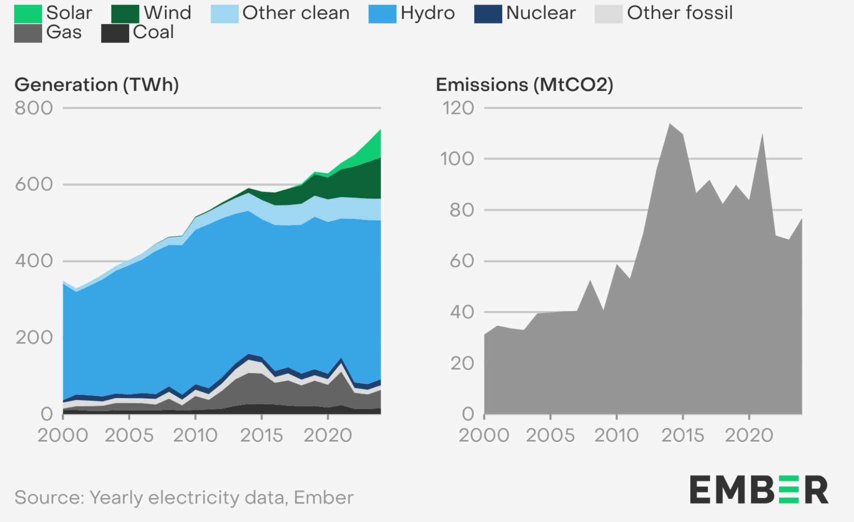
Brazil's demand increase was primarily met by solar (+23 TWh, +45%) and wind (+12 TWh, +13%). Its increase in solar generation was the third-largest of any country in 2024.



Brazil became the fifth-largest solar generator in 2024, surpassing Germany in the global rankings. Brazil also had the third-largest increase in wind generation in 2024, maintaining its fourth place globally in terms of total annual wind generation.

Hydro generation was down 14 TWh (-3.2%) in 2024 due to an intense and widespread drought. Brazil's fossil generation increased by 12 TWh (+18%), primarily due to gas increasing by 10 TWh (+27%), countering the fall in hydro generation.

Brazil has a high share of renewables due to its large hydroelectric base and the rapid expansion of solar and wind power in recent years. Its share of wind and solar has been growing rapidly in recent years, reaching 24% in 2024, a substantial increase from 17% in 2022 and up from just 5.8% in 2016.



Brazil's power sector emissions peaked in 2014 at 114 million tonnes of CO_2 ($MtCO_2$). Growing wind and solar have since reduced the need for fossil power. In 2024, a decade past the peak, its power sector emissions were 32% below 2014 levels, at 77 $MtCO_2$.

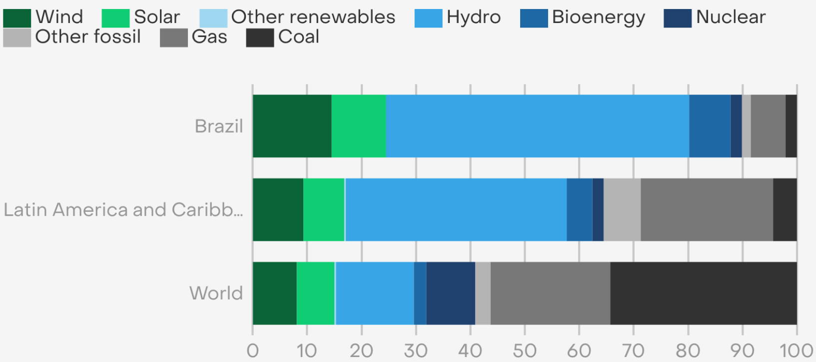
Brazil's power sector emissions have fluctuated since the peak due to variability in weather conditions, with rainfall and drought affecting hydro output. In 2024, drier conditions prevailed and overall, there was a slight decline in hydro generation (-14 TWh) that was almost all offset by rising fossil generation (+12 TWh) such that power sector emissions increased slightly from 2023 to 2024 (+8 $MtCO_2$).

Brazil generated 90% of its electricity from clean sources in 2024, with hydro dominating the mix at 56%. Its share of wind and solar (24%) was above the global average (15%) and the average in Latin America (17%).

Over 87% of Brazil's electricity came from renewables in 2024, which was by far the highest of any G20 country and almost triple the global average of 32%.

Brazil: Electricity mix compared to the region and world

Share of electricity generation, by source (%)



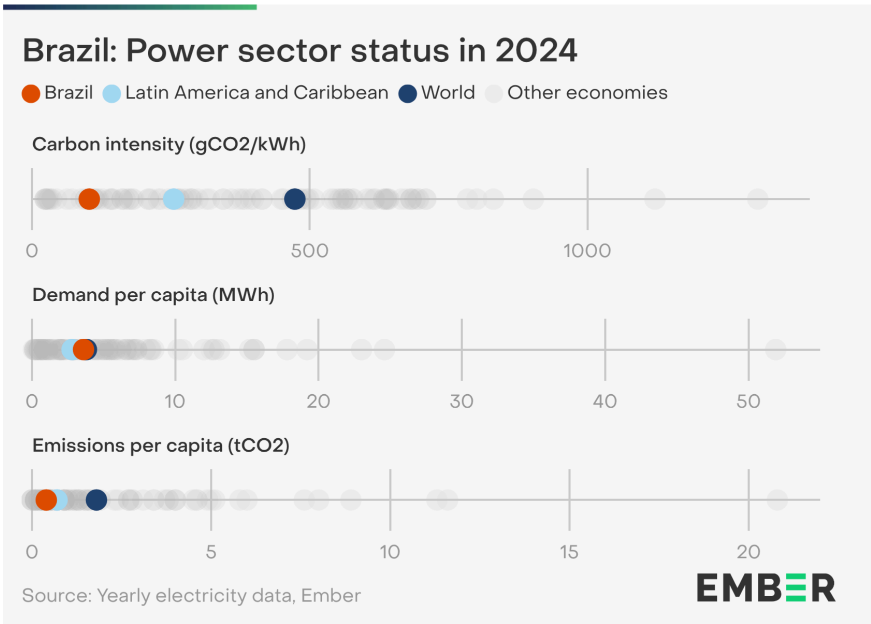
Source: Yearly electricity data, Ember

Brazil relied on fossil fuels for just 10% of its electricity in 2024, a sixth of the global average of 59%.

In 2024, Brazil's carbon intensity of electricity generation was 103 gCO_2/kWh, the second-lowest in the G20 and less than a quarter of the global average (473 gCO_2/kWh).

Brazil's electricity demand per capita was 3.6 MWh in 2024, just below the world average of 3.8 MWh but above the regional average of 2.8 MWh.

Brazil's per capita emissions from electricity generation (0.4 tCO2) are the lowest in the G20 and less than a quarter of the global average of 1.8 tCO2.



Clean power poised to begin the era of fossil fuel decline

In a world of higher electricity demand growth, clean electricity is stepping up to the challenge. Spearheaded by exponential solar expansion, clean power is set to grow faster than demand, marking the start of a permanent decline in fossil generation.

2024 both clarified and consolidated the shape of the global clean power transition. At first glance it might appear an unremarkable year, as global fossil generation increased for another year. However, the main driver of this increase was the change in temperature between 2024 and 2023. Stripping out these temperature effects, clean generation met 96% of the demand increase in 2024. Isolating the trend from the short-term fluctuations, it becomes clear that the world is very close to an era of falling fossil fuel generation.

2024 also cemented solar power as the engine of the global energy transition. Global solar generation increased by nearly a third, continuing its lead as the largest and fastest-growing source of new electricity. Its rise was on display everywhere in 2024, from world leaders like China, India, Brazil and the EU to new markets in Saudi Arabia and Pakistan. The startling expansion and cost reduction of battery storage offer another positive, enhancing the economics of solar power still further.

New drivers of demand – data centres for AI and cryptomining, and electrified transport and heating – raise pressing questions about how to meet rising power consumption. 2024 has shown that solar and batteries, the new engines of growth, are best placed to deliver cheaply and at scale, and are already doing so. While demand growth is undoubtedly going to be higher than in previous years, Ember's analysis shows that clean growth is set to outpace it, making long-term investments to grow fossil generation a risky bet.

Any near-term increases in fossil fuel generation should not be mistaken for failure of the energy transition. As we pass the tipping point where clean generation structurally outpaces demand growth, any changes to fossil fuel generation over the short-term will mostly reflect fluctuations in weather, as seen in 2024 with the impacts of heatwaves. But while changes in fossil generation in the short-term may be noisy, the direction and ultimate destination are unmistakable. The global energy transition is no longer a question of if, but how fast.

The significant advances of 2024 were achieved despite unfavourable geopolitical headwinds. In 2025 those winds are strengthening again, with an increased prospect of tariffs on international trade and a US administration pinning its energy hopes on fossil fuels. However, while playing an important role, the US is not the central driver of the global clean energy transition – this role has fallen more and more to China. China registered more than half of the global increase in both solar and wind power in 2024 and is the world leader in both clean energy manufacturing and deployment. There is no indication that China is slowing its own transition to an electrified economy, nor that its trade relationships with other Global South countries will do anything but intensify.

Beyond China, the rapid clean energy build registered by India, Brazil and the EU is a sign that other major economies are forging ahead. Notably, India is rapidly expanding its own solar and battery manufacturing sector. Meanwhile, uncertainties over trade relations with the US are likely to increase many governments' appetite for the clean energy transition, in regions including Europe and Latin America, because it offers a route to reduced dependence on volatile global markets for coal and gas.

The imperative for the energy transition is clear: the world is upgrading from an inefficient energy system reliant on securing a constant supply of expensive and polluting fossil fuels. In the new energy system, electricity will be at the heart, with clean electrons powering everything from transport to steel production. Solar and wind, backed by a suite of clean flexibility solutions like improved grids and storage, will be the engine powering the world. 2024 shows unmistakably that the transition to this new energy system is very much underway. Governments and businesses that act decisively to embrace clean energy and quickly move away from fossil fuels will be rewarded with a more resilient, competitive, and secure energy future.

Methodology

Generation, imports and demand

Yearly data from 2000 to 2023 is gross generation, taken primarily from the Energy Institute's Statistical Review of World Energy, the Energy Information Administration (EIA), Eurostat and IRENA. 2024 data is an estimate of gross generation, based on generation gathered from monthly data. This estimate is calculated by applying absolute changes in monthly generation to the most recent annual baseline.

Net imports from 1990 to 2023 are taken from the EIA and Eurostat, with recent data estimated in the same manner as generation. Demand is calculated as the sum of generation and net imports, and where possible validated against published direct demand figures. Because it uses gross generation and does not include transmission and distribution losses, it will tend to be higher than end-user demand.

Monthly data is gathered for 88 countries from over 70 sources, including national transmission system operators and statistical agencies, as well as data aggregators such as ENTSO-E. In some cases, data is published on a monthly lag; here we have estimated recent months based on our generation model.

Monthly data is often reported provisionally, and is far from perfect. Every effort has been made to ensure accuracy, and where possible we compare multiple sources to confirm their agreement.

Bioenergy has typically been assumed (by the IPCC, the IEA, and many others) to be a renewable energy source, in that forest and energy crops can be regrown and replenished, unlike fossil fuels. It is included in many governmental climate targets, including EU renewable energy legislation, and so Ember includes it in “renewable” to allow easy comparison with legislated targets. However, we recognise the IPCC reported lifecycle carbon intensity of bioenergy is significantly higher than other renewables and nuclear, and this is incorporated into our power sector emissions estimate. More information about Ember’s classification of electricity sources can be found in the full methodology for Ember’s Yearly Electricity Data under “Fuel Types”.

References to CO2 emissions in this report are using CO2 equivalent emissions which include emissions from other greenhouse gases such as methane (CH4). Power sector emissions are based on the methodology from Ember’s Yearly Electricity Data which can be found here.

Calculation of temperature impacts on electricity demand

This methodology outlines the approach used to quantify the impact of temperature on global electricity demand. By using regression analysis with temperature data and monthly electricity demand, this analysis isolates temperature-driven variations from structural changes in electricity consumption.

Geographic coverage

The analysis aims to deliver a global level account of temperature effects on demand. For this purpose, it includes country/region-level analysis for 14 countries and regions. The analysis was carried out for Australia, Brazil, Canada, China, the EU, India, Iran, Japan, Mexico, Russia, South Africa, South Korea, Türkiye, the United Kingdom and the United States. Together, these power sectors accounted for 82% of global electricity demand. The assessed impact in included countries is scaled to match 100% of global demand.

Data Sources

Meteorological Data: Temperature data was sourced from the ERA5 reanalysis via Open-Meteo, providing hourly 2-metre temperature.

Electricity Demand Data: Monthly electricity demand data was obtained from Ember’s Monthly Electricity data.

Population Data: Population figures were derived from NASA's Gridded Population of the World dataset, with a resolution of 0.25 degrees (~30 km).

Time Period: The analysis of temperature effects covers the period from 2015 to 2024, depending on the availability of data for each country.

Normalising Monthly Electricity Demand

Monthly electricity demand was normalised by dividing observed demand by its 12-month trailing average. This process helps to minimise the influence of long-term structural factors such as economic growth or increased electrification, isolating seasonal and temperature-related changes.

Temperature Data

A 1-degree resolution grid was created for each country. Population data was assigned to each grid cell, and hourly temperature data was extracted for the centre of each cell.

Population Weighting for temperature and CDD/HDD values

Population weighted temperatures were calculated by weighting temperature for each grid cell by the corresponding population.

Similarly, cooling degree days (CDD) and heating degree days (HDD) values were calculated and weighted using the same approach. This ensures that temperature metrics reflect conditions in populated areas, where electricity demand is concentrated. It also accounts for the non-linear relationship between temperature and energy use. Cooling degree days and Heating degree days were defined as follows:

- Cooling degree days (CDD): The sum of degrees by which daily temperatures exceed 22°C, reflecting cooling demand.
- Heating degree days (HDD): The sum of degrees by which daily temperatures fall below 18°C, reflecting heating demand.

Regression Analysis

To understand the relationship between temperature and electricity demand, we used regression analysis to identify how changes in temperature, measured through cooling degree days and heating degree days, influence electricity demand per country. The analysis helps quantify the extent to which higher or lower temperatures drive changes in monthly electricity use.

Calculating Anomalies
Monthly temperature, HDD and CDD anomalies were calculated with respect to a ten-year baseline (2015-2024). Using a relatively short and recent baseline makes the anomalies more relevant to recent changes in electricity demand.

Calculating Absolute Impact
The impact of temperature on demand was derived by applying the identified relationship of CDD and HDD data and monthly electricity demand to the CDD and HDD anomalies. These were then scaled by normalised demand into absolute impacts in TWh. For some countries, monthly demand reporting can be lower than annual reported values. Where applicable, temperature impact values derived from monthly data were scaled to match annual demand figures. Combined country level assessments for the 15 included regions were aggregated and scaled to match total global electricity demand of all countries in 2024.

Demand disaggregation

Electricity demand growth from EVs is estimated from changes in EV stock by vehicle type, multiplied by reference values for electricity consumption by vehicle type. Vehicle types include passenger cars, buses, trucks and vans, for both battery electric vehicles and plug-in hybrid electric vehicles. Historic stock data is taken from the IEA's Global EV Data Explorer. Demand growth in 2024 is estimated using sales data for passenger cars taken from BNEF, and assumes recent growth rates are maintained across other vehicle types.

Electricity demand growth from data centres (including cryptocurrency mining) is estimated using IEA data. As IEA data only goes back to 2019, the same value for demand growth was used in 2019 as in 2020, based on the assumption that growth at this point in time was linear (see assessment of historic estimates from Lawrence Berkeley National Laboratory). For 2023-2024, IEA projections for cryptocurrency mining have been substituted for real data from Cambridge University's Cambridge Bitcoin Electricity Consumption Index (CBECI) to more accurately estimate cryptocurrency mining electricity consumption growth for recent years. The estimates for cryptocurrency mining electricity demand growth are estimated using growth in Bitcoin electricity demand, scaled to fit estimates for total crypto demand in

2022. Bitcoin makes up the large majority of proof-of-work cryptocurrency mining activity and other cryptocurrencies based on proof-of-stake mechanisms use considerably less energy.

Electricity demand growth from heat pumps is estimated using changes in stock data, multiplied by annual average consumption. Data for 2024 assumes that the reduction in sales in key markets seen in the first half of 2024 continued throughout the rest of the year.

Other data sources

This report makes use of a variety of datasets curated by Ember, including data on exports of Chinese solar PV modules and global solar capacity installations.

A full methodology for Ember's data on exports of Chinese solar PV modules can be found here.

Solar capacity additions are estimated using available monthly capacity data from national sources for countries that made up 80% of solar capacity additions in 2023. National monthly data is generally available for all of 2024, with exceptions for a few countries where the final months of 2024 are estimates based on market growth rates in available months. Estimates for remaining countries are made through analysis of Chinese solar PV module export data. A full methodology can be found here.

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Cover image
The cover of this report was designed by Reynaldo Dizon and is based on real data used for the analysis within this report, showing growth in solar power across the world – as a share of generation, and in absolute terms – over the past five years.